FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS — GRADE 10 | SUB-STRAND 3.2: PROBABILITY I	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

YOUR TASK — THE FINAL EXPLANATION CHALLENGE

You have spent 8 lessons investigating one big question: "How can Kamau — and other Kenyans facing uncertain everyday outcomes — use the mathematics of probability to make smarter, fairer, and safer decisions?"

You started at Machakos Country Bus Station, watching Kamau puzzle over his tally notebook. You now have the mathematical tools to answer what confused him. This is your moment to bring everything together.

Write a complete, well-organised Final Explanation that directly answers the driving question. Use Kamau's matatu data AND at least one other Kenyan everyday context you have studied or can think of. Your explanation must show — not just mention — the mathematics. Show your working, use correct vocabulary, and explain what your numbers actually mean for real decision-making.

READ ALL FOUR SECTIONS AND THE RUBRIC BEFORE YOU START WRITING. Each section builds on the one before it, just like the unit did.

Section 1 — Why Uncertainty Exists (and Why That Is Not a Problem)	
Explain why Kamau cannot predict with certainty whether the next seat on his matatu will be filled, even after recording 20 trips. Use the words OUTCOME, SAMPLE SPACE, and EXPERIMENTAL PROBABILITY in your explanation. Then calculate the experimental probability that a	Even though Kamau records every trip carefully, he cannot be certain about the next trip because each passenger makes an independent decision about whether to board — no single passenger's choice forces another person to travel. This means the outcome of each trip is uncertain. The sample space for a single seat is simple — it is either filled (F) or empty (E) — but across all 14 seats and many trips, the combinations are enormous, making a perfect prediction impossible. Using Kamau's 20 recorded trips, I can count the fully-booked trips (passenger count = 14): trips 3, 5, 7, 10, 12, 15, 17, 19 ... let me list them carefully from the data: 6, 9, 14✓, 11, 14✓, 8, 14✓, 12, 7, 14✓, 10, 14✓, 13, 6, 14✓, 9, 14✓, 11, 8, 14✓. That gives 8 fully-booked trips out of 20.

trip is fully booked (all 14 seats filled) using Kamau's tally notebook data: trips recorded were 6, 9, 14, 11, 14, 8, 14, 12, 7, 14, 10, 14, 13, 6, 14, 9, 14, 11, 8, 14. Finally, explain what this probability tells Kamau — and what it does NOT tell him about tomorrow's trip.	Experimental probability of a full matatu = Number of fully-booked trips ÷ Total trips recorded = 8/20 = 2/5 = 0.4 = 40%. This tells Kamau that, based on past experience, roughly 4 in every 10 trips have been fully booked. It is a useful estimate for planning — for example, he can expect about 40% of his weekly trips to earn maximum revenue. However, it does NOT tell him that tomorrow's specific trip will be full. Experimental probability describes a long-run pattern, not a guarantee for one event. The more trips Kamau records, the more reliable his estimate becomes — this is the Law of Large Numbers at work.

Section 2 — Mutually Exclusive and Independent Events in Kamau's World	
Kamau notices two things about his passengers: (a) on any single trip, the matatu cannot be both fully booked AND have empty seats at the same time; and (b) whether Passenger A boards at Machakos Bus Station does not change the probability that Passenger B — a stranger — will board at the next stop in Mlolongo. Using these observations, define MUTUALLY EXCLUSIVE EVENTS and INDEPENDENT EVENTS with correct mathematical notation. Then apply the Addition Law for mutually exclusive events AND the Multiplication Law for independent events using numbers from Kamau's data or a realistic Kenyan context you choose. Show all working and explain what the results mean.	MUTUALLY EXCLUSIVE EVENTS — Two events are mutually exclusive if they cannot both happen at the same time; P(A and B) = 0, and therefore P(A or B) = P(A) + P(B). For Kamau: Let F = 'the matatu is fully booked' and E = 'the matatu has at least one empty seat'. These are mutually exclusive because a matatu with 14 seats cannot be simultaneously full and have an empty seat. From the data: P(F) = 8/20 = 2/5. The remaining 12 trips had empty seats, so P(E) = 12/20 = 3/5. Check: P(F or E) = P(F) + P(E) = 2/5 + 3/5 = 5/5 = 1. ✓ This makes sense — every trip must be one or the other. The probability of 1 confirms these two events cover all possible outcomes (they are also exhaustive). INDEPENDENT EVENTS — Two events are independent if the occurrence of one does not affect the probability of the other; P(A and B) = P(A) × P(B). For Kamau: A market trader from Kitengela deciding to board at Machakos Bus Station and a student from Athi River deciding to board at Mlolongo are strangers making separate decisions. Suppose P(trader boards) = 0.4 and P(student boards) = 0.6. P(both board on the same trip) = P(trader) × P(student) = 0.4 × 0.6 = 0.24 = 24%. This means there is roughly a 1-in-4 chance that both specific passengers appear on the same trip — useful if Kamau is reserving two seats for regular customers. Because the events are independent, knowing the trader boarded gives Kamau no new information about whether the student will board.

Section 3 — Tree Diagrams: Modelling Two Consecutive Stops	
Kamau's matatu stops at Machakos Bus Station first, then at Mlolongo. At each stop, a seat is either filled (F) or	TREE DIAGRAM (described in text since this is a written response): First branch — Machakos Bus Station: P(F_1) = 3/5 and P(E_1) = 2/5.

empty (E). Suppose the probability a seat is filled at Machakos Bus Station = 3/5, and the probability a seat is filled at Mlolongo = 1/2 (these are independent events). Draw a fully labelled tree diagram showing all possible outcomes across the two stops. List the complete sample space. Then calculate: (i) the probability the seat is filled at both stops; (ii) the probability the seat is filled at exactly one of the two stops; (iii) the probability the seat is empty at both stops. Show all working and explain what Kamau could do with this information.	Second branch from F_1 — Mlolongo: $P(F_2) = 1/2$ and $P(E_2) = 1/2$. Second branch from E_1 — Mlolongo: $P(F_2) = 1/2$ and $P(E_2) = 1/2$. COMPLETE SAMPLE SPACE = $\{F_1F_2, F_1E_2, E_1F_2, E_1E_2\}$ CALCULATIONS: (i) P(filled at both stops) = $P(F_1F_2) = P(F_1) \times P(F_2) = 3/5 \times 1/2 = 3/10 = 0.3 = 30\%$. (ii) P(filled at exactly one stop) = $P(F_1E_2) + P(E_1F_2)$ $= [P(F_1) \times P(E_2)] + [P(E_1) \times P(F_2)]$ $= [3/5 \times 1/2] + [2/5 \times 1/2]$ $= 3/10 + 2/10 = 5/10 = 1/2 = 50\%$. (iii) P(empty at both stops) = $P(E_1E_2) = P(E_1) \times P(E_2) = 2/5 \times 1/2 = 2/10 = 1/5 = 20\%$. VERIFICATION: $3/10 + 5/10 + 2/10 = 10/10 = 1$ ✓ What Kamau can do with this: There is a 20% chance a seat earns nothing across both stops — that is 1 in 5 seats losing money on that section of the route. If his matatu has 14 seats, he can estimate roughly $14 \times 0.2 \approx 3$ seats may be empty across both stops, helping him decide whether to adjust departure timing or add a stop closer to a market or school.

Section 4 — Beyond the Matatu: Probability for Smarter, Fairer, and Safer Kenyan Decisions	
The driving question asks about Kenyans facing uncertain everyday outcomes — not just Kamau. Choose TWO different real-life Kenyan situations (other than the matatu) where probability can support better decision-making. For each situation: name the context and who benefits, identify at least one event and write its probability (estimated or calculated), state whether events involved are mutually exclusive, independent, or neither — and justify your answer, and explain how knowing this probability leads to a smarter, fairer, OR safer	SITUATION 1 — A maize farmer in Eldoret planning for crop disease. A farmer in Uasin Gishu County knows from county agricultural records that in any given season, the probability that her maize crop is affected by fall armyworm is 0.35, and the probability it is affected by drought stress is 0.25. She wants to know the probability that at least one of these affects her crop. Are these events mutually exclusive? No — her maize can suffer from both armyworm AND drought in the same season, so P(armyworm AND drought) ≠ 0. Using the General Addition Law: P(armyworm or drought) = P(armyworm) + P(drought) − P(both) $= 0.35 + 0.25 - (0.35 \times 0.25) = 0.60 - 0.0875 = 0.5125 \approx 51\%$. Decision benefit: There is roughly a 51% chance the farmer faces at least one major threat each season. This means she should budget for pesticide or irrigation in more than half of her seasons — a fairer, evidence-based plan rather than guessing. It also helps the county government prioritise which farmers receive subsidised inputs. SITUATION 2 — A health worker in Kisumu tracking malaria risk. A community health worker knows that the probability a child under 5 in a lakeshore village near Lake Victoria tests positive for malaria in a given month is 0.18. She tests two children from different households (independent events).

decision. Then write a closing paragraph that directly and fully answers the driving question in your own words.	P(both children test positive) = 0.18 × 0.18 = 0.0324 ≈ 3.2%. P(at least one tests positive) = 1 − P(neither tests positive) = 1 − (0.82 × 0.82) = 1 − 0.6724 = 0.3276 ≈ 33%. Decision benefit: A 33% chance that at least one of two randomly selected children tests positive is high enough to justify keeping rapid test kits and treatment at the village clinic at all times — a potentially life-saving, evidence-based resource allocation decision. CLOSING PARAGRAPH — ANSWERING THE DRIVING QUESTION: Kamau cannot predict with certainty whether tomorrow's matatu will be full, and the Eldoret farmer cannot guarantee her maize will survive, and the Kisumu health worker cannot know in advance which child will fall ill. But uncertainty does not mean helplessness. By collecting data and calculating experimental probability, listing sample spaces, applying the laws of probability for mutually exclusive and independent events, and building tree diagrams to model sequences of uncertain outcomes, all of these Kenyans can replace guesswork with mathematical reasoning. Probability does not remove uncertainty — it gives us a rigorous, honest way to measure it, communicate it, and act wisely in spite of it. That is how mathematics makes everyday Kenyan decisions smarter, fairer, and safer.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of Probability Concepts	Accurately defines and applies experimental probability, theoretical probability, mutually exclusive events, independent events, and sample space. Uses correct mathematical notation (e.g. $P(A \cup B)$, $P(A \cap B)$) throughout. Explanations show deep conceptual understanding — student can articulate what probability does and does not tell us.	Correctly defines and applies most key concepts with minor errors in notation or explanation. Understands the difference between experimental and theoretical probability. May have small gaps but the core understanding is clear.	Shows partial understanding of one or two concepts (e.g. can calculate experimental probability but confuses mutually exclusive with independent events). Definitions are incomplete or contain errors. Mathematical notation is missing or incorrect.
Mathematical Accuracy and Working	All calculations are correct and fully shown step by step. Fractions, decimals, and percentages are used accurately and interchangeably where appropriate. Verification steps (e.g. checking all probabilities sum to 1) are included. Tree diagram is complete and correctly labelled with all branches and products shown.	Most calculations are correct with working shown. Minor arithmetic errors do not undermine the overall solution. Tree diagram is present and mostly correct. At least one verification step is attempted.	Calculations contain significant errors or working is missing, making it impossible to follow the student's reasoning. Tree diagram is incomplete, missing labels, or contains incorrect probabilities. Results are not verified.
Use of Kenyan Real-Life Contexts	Kamau's matatu data is used accurately and meaningfully throughout Section 1–3. Two additional Kenyan contexts in Section 4 are specific, realistic, and well-chosen (e.g. named counties, crops,	Kamau's data is used correctly. At least one additional Kenyan context is clearly described and connected to probability. The link between the mathematics and the decision is stated but may lack detail	Kamau's data is used incorrectly or ignored. Additional contexts are vague, non-Kenyan, or not connected to probability calculations. The real-world decision-making link is absent or unclear.

	health settings). The connection between the mathematics and the real decision is clearly explained — the student shows who benefits and how.	about who benefits or how the decision improves.	
Communication and Mathematical Language	All required vocabulary (outcome, sample space, experimental probability, mutually exclusive, independent, Addition Law, Multiplication Law) is used correctly and consistently. Writing is well-organised with clear section structure. The closing paragraph directly, fully, and confidently answers the driving question in the student's own words.	Most required vocabulary is used correctly. The response is mostly well-organised with a clear attempt to answer the driving question in the closing paragraph. Minor lapses in language precision do not obscure meaning.	Key vocabulary is missing, misused, or used without explanation. The response lacks clear organisation. The closing paragraph is absent, too brief, or does not address the driving question.
Connections Across the Unit (Synthesis)	The student explicitly connects at least four of the following unit concepts and shows how they build on each other: experimental probability → theoretical probability → sample space → mutually exclusive events → independent events → tree diagrams → Addition and Multiplication Laws. The response reads as a unified, cumulative argument, not a list of separate topics.	The student connects at least three unit concepts and shows some understanding of how they relate. The response has a logical flow but may treat some topics in isolation rather than showing how they build toward answering the driving question.	The student addresses concepts separately without connecting them. The response reads as a list of definitions or calculations with no overall argument. Fewer than three unit concepts are addressed.